Multimedia Signal Processing - 2nd module

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1 Question - 10 pts

Consider a source X that emits symbols from a binary alphabet a, b. Let $p(i) = Pr\{X = i\}$, $p(i|j) = Pr\{X_k = i|X_{k-1} = j\}$. Encode the sequence of symbols $aaba$ by means of Elias coding, showing the corresponding bitstream. Consider the following three cases:

1. Assume that X is a discrete memoryless source, with $p(a) = 0.8$
2. Assume that X is a discrete source with memory, with $p(a) = 0.8$, and
   - $p(a|a) = 0.9$
   - $p(a|b) = 0.3$
3. Assume that X is a discrete source with memory, with $p(a) = 0.8$. Conditional probabilities are to be estimated as encoding progresses.

2 Question - 5 pts

Illustrate the block diagram of a lossy image coding architecture based on transform coding (both encoder and decoder).

3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X denote a continuous source, and $D(R)$ the corresponding rate-distortion function, and σ_x^2 its variance.
   - [T] [F] $H(X) = \frac{1}{2} \log \frac{\sigma_x^2}{D}$
   - [T] [F] $R_1 > R_2 \Rightarrow D(R_1) \geq D(R_2)$
   - [T] [F] $D(R) = \epsilon \sigma_x^2 2^{-2R}$
2. Consider a speech coding system
   - [T] [F] Coding intelligible speech typically requires at least 32 kbps
   - [T] [F] The perceptual filter $W(z)$ weights more the error in the formant regions of the spectrum
   - [T] [F] Voiced speech can be synthesized based on the spectral envelope only
   - [T] [F] The computational complexity of the encoder in DPCM is smaller than in analysis-by-synthesis vocoders
3. Consider a video coding system and a GOP structure $IPP...I$
   - [T] [F] The coding efficiency improves by increasing the GOP size
   - [T] [F] The computational complexity increases by increasing the GOP size
   - [T] [F] The adoption of P-frames introduces drift

4 Question - 12 pts

Load the signal x (`load speech`), which contains a speech fragment quantized at 8 bits per sample.

1. Consider the signal as a memoryless source that emits symbols in [-128:127]. Which is the minimum rate achievable if the signal is coded lossless?
2. Normalize x between -1 and 1 and compare the performance of PCM and DPCM lossy coding. Plot the rate-distortion curves for the two coding schemes. Let the number of bits per sample vary between 1 and 7 and express the rate in kbps.